

vehicle to provide a complete 360° view of its immediate surroundings, making it ideal for automotive advanced driver assistance systems and autonomous



vehicle applications such as blind spot detection, small object detection, automated park assist, and free space estimation around the vehicle. It is also suited for accurate detection of objects in proximity such as small children, on-road objects, road edges, and stationary objects.

www.businesswire.com

Coreless current sensor

Allegro MicroSystems has launched a new coreless Hall-effect current sensor for demanding automotive and industrial systems, offering higher accuracy and sensitivity, additional fault detection capabilities, and user programmability. The new ACS37610 sensor is designed to



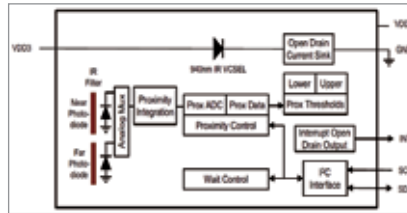
measure currents from 100A to greater than 4,000A flowing through a busbar or PCB trace with a typical 1% accuracy—with no need for an external concentrator or U-shaped magnetic shield. It is an optimal choice for electric vehicle high-voltage traction motor inverters, 48V/12V auxiliary inverters, heterogeneous redundant battery monitoring, over-current detection, smart fuses, power distribution units, and power supplies. It enables system designers to achieve enhanced efficiency and higher power density while reducing system complexity, BoM, cost, footprint, and weight.

<https://allegromicro.com>

Proximity sensor for wireless earbuds

ams has launched the TMD2636, a fully integrated proximity sensor that

occupies 30% less space than currently available solutions and so delivers transformational added value to manufacturers of True Wireless Stereo earbuds.



Reliable proximity detection performed by one or more TMD2636 sensors enables the earbud to automatically power up when inserted in the ear and power down when removed. This maximises power savings when the earbud is not in use while providing the best user experience. The sensor module integrates a 940nm infrared (IR) vertical-cavity surface-emitting laser emitter, IR-sensing photodiodes, sophisticated control circuitry, and a clear-mould overlay in a 2.0mm × 1.0mm × 0.35mm package.

<https://ams.com>

Solid-state lidar multibeam scanner

The scanGrid2 safe multibeam scanner from SICK is a compact sensor that uses a novel and in-house developed solid-state lidar technology to increase the productivity of small autonomous and line-guided transport vehicles. Certified as a Type 2/SIL 1 safety sensor,



it can protect hazardous areas up to performance level c and works well for collision avoidance. An app and cloning function also ensure a high level of up time and fast fleet deployment. The scanGrid2 safely detects objects of a variety of sizes within the freely configurable protective field zones, can evaluate multiple fields, and can execute configurable monitoring cases. The solid-state lidar technology is based on the principle of time-of-flight measurement and eliminates all moving parts. Instead, the device uses only semiconductor elements in conjunction with geometrically arranged optics modules

to span a protective field of 150 degrees. www.sick.com

Multifunction RFIC chip

Vayyar Imaging has launched a multi-range 'XRR' chip, a single RFIC with a range of 0-300m, designed for passenger cars, trucks, and motorcycles. Supported by a 48-antenna MIMO array, the leading-edge platform provides radar imaging with unprecedented accuracy for numerous safety applications, without the need for external processors. The chip



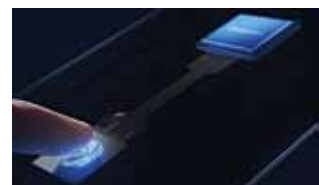
is unmatched in terms of range, resolution, and performance. With an ultra-wide field of view and rich 4D point cloud

imaging, it delivers unprecedented multifunctionality on a single-chip platform, supporting dozens of advanced driver assistance systems, advanced rider assistance systems, and autonomy features. This eliminates the need for multiple, costly vehicle sensors and reduces costs, complexity, hardware, software, power consumption, wiring, and integration efforts.

<https://blog.vayyar.com>

Fingerprint scanning sensor

Qualcomm 3D Sonic Sensor Gen 2 delivers a better user experience by using technological advances and ultrasonic waves to scan 3D features of a user's finger for a deeply accurate image, allowing to scan fingerprints through



solid surfaces like glass and metal even while wet. The sensor is ultra-thin, measuring 0.2mm, which enables OEMs to build mobile devices with cutting-edge form factors including full edge-to-edge flexible OLED displays. The sensor provides a larger region for you to place your finger and captures 1.7x more biometric data with 50% faster processing. www.qualcomm.com